Code No: 126AP **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, May - 2019
DISTRIBUTED SYSTEMS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

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PART - A

(25 Marks)

1.a)	List down the examples of distributed systems.	[2]
b)	Present a note on peer-to-peer architecture.	[3]
c)	Differentiate synchronous and asynchronous systems.	[2]
d)	Discuss about Lamport distributed mutual exclusion algorithm.	[3]
e)	Discuss about client-server communication.	[2]
f)	What is interprocess communication? Explain.	[3]
g)	List down the characteristics of file systems.	[2]
h)	Write a short note on directory service interface.	[3]
i)	Give a brief idea about transactions in distributed systems.	[2]
j)	Write about flat transactions.	[3]

PART - B

(50 Marks)

2.a) What are the challenges of resource sharing and web?
b) Differentiate fundamental and architectural models. [5+5]

OR

3. Explain the following.
a) Intranet
b) Mobile Computing
c) Secure Channels. [3+3+4]

4.a) Explain about distributed debugging.
b) Discuss about time and global states in distributed systems. [5+5]

OR

5.a) What is multicast communication? Explain.
b) Explain about synchronizing physical clocks. [5+5]

6. Explain in detail about external data representation and Marshalling. [10]

OR

7.a) Explain about events and notifications.
b) Discuss about JAVA RMI. [5+5]

8.a) Explain about name services and the domain name Systems.
b) Explain about sun network file system. [5+5]

OR

9.a) Explain about global name services.
b) Discuss about File service architecture. [5+5]

10.a) Discuss about locks in distributed systems.
b) What are the different comparison methods for concurrency control? [5+5]

OR

11.a) Explain about transaction recovery in distributed systems.
b) Explain about distributed deadlocks. [5+5]

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